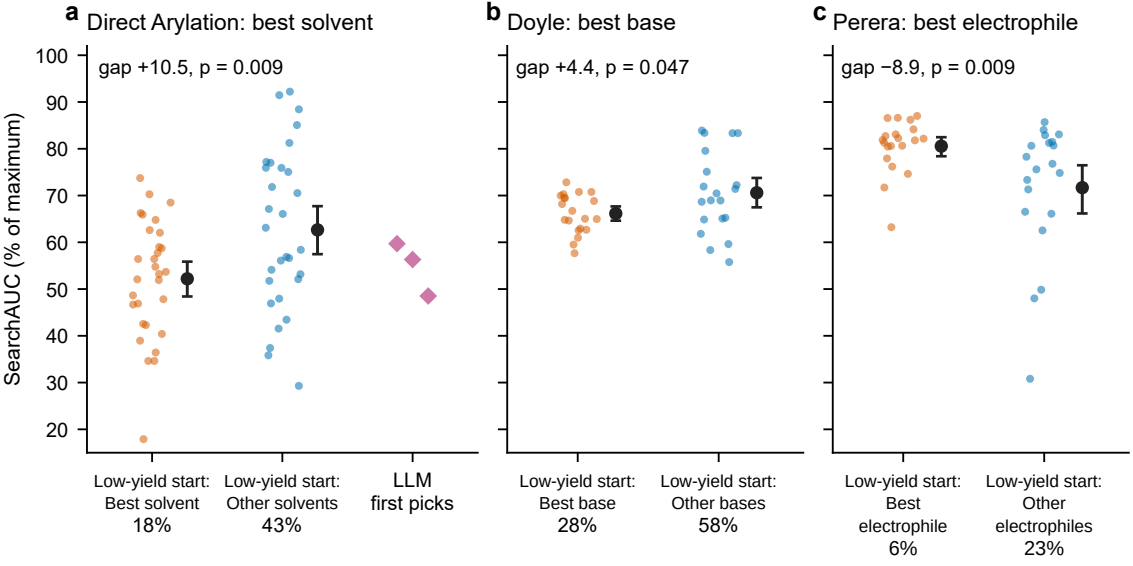
a
Direct Arylation: best solvent
gap +10.5, p = 0.009
b
Doyle: best base
gap +4.4, p = 0.047
c
Perera: best electrophile
gap −8.9, p = 0.009
SearchAUC (% of maximum)
100
90
80
70
60
50
40
30
20
Low-yield start: Best solvent 18%
Low-yield start: Other solvents 43%
LLM first picks
Low-yield start: Best base 28%
Low-yield start: Other bases 58%
Low-yield start: Best electrophile 6%
Low-yield start: Other electrophiles 23%